

Scalability & Cost Trade-off Analysis

Cost Analysis — per design

Design	Complexity	Added overhead	Key saving
D1 Adaptive Dispatch	—	CPU O(nnz) scan, hidden under H2D	matrix-aware merge/hash pick, ~0 GPU cost
D2 Column-Domain Merge	O(flops), 3 axes	column-bucket binning + sync	removes global sort (54-70% of ESC)
D3 Light-Symbolic MinHash	O(nnz) sizing	per-row sketch + 1.5x safety margin	1.4-3.3x faster symbolic (vs O(flops))
D4 Tail-Balanced A^T A	~ -50% arithmetic (upper triangle)	CPU O(nnz) workload + reshape	kills long-tail straggler
Overall SpGEMM	compute-bound	accumulation / dedup = the floor	engineering opts saturate here

Scalability — runtime vs. size

